

LECTURE 1 - SECTION 6.1. Risk sharing - DETAILS OF THE ALGEBRA -

$$c_i^i(s^t) = c_i^j(s^t) \left(\frac{\mu_i}{\mu_j} \right)^{-\frac{1}{\sigma}} \quad (*)$$

$$\sum_i c_i^i(s^t) = \sum_i y_i^i(s^t) = \bar{y}_t(s^t)$$

Assume $c_i^i = d_i \bar{y}_t$, $\sum d_i = 1$

$$(\dagger) \quad d_i \bar{y}_t = d_j \bar{y}_t \left(\frac{\mu_i}{\mu_j} \right)^{-\frac{1}{\sigma}}$$

$$d_i = d_j \left(\frac{\mu_i}{\mu_j} \right)^{-\frac{1}{\sigma}}$$

$$d_j = \left(\frac{\mu_i}{\mu_j} \right)^{\frac{1}{\sigma}} d_i \Rightarrow \sum_j d_j = 1 = \sum_j \left(\frac{\mu_i}{\mu_j} \right)^{\frac{1}{\sigma}} d_i \Rightarrow$$

$$d_i = \left(\sum_j \left(\frac{\mu_i}{\mu_j} \right)^{\frac{1}{\sigma}} \right)^{-1}$$

i "poor" $\Rightarrow \mu_i$ large $\Rightarrow d_i$ small



price formula: $\beta^r u'(c_r^i(s^r)) \pi_r(s^r) = \mu_i^{-1} q_r^0(s^r)$

with CRRA: $\beta^r \left[d_i \bar{y}_r(s^r) \right]^{-\sigma} \pi_r(s^r) = \mu_i^{-1} q_r^0(s^r)$

$$\Rightarrow q_i^0(s^r) = \mu_i^{-1} d_i^{-\sigma} \beta^r (\bar{y}_r(s^r))^\sigma \pi_r(s^r)$$

Use the normalization: $\mu_i^{-1} d_i^{-\sigma} = 1 \Rightarrow q_i^0(s^r) = \beta^r (\bar{y}_r(s^r))^{-\sigma} \pi_r(s^r)$

How to compute d_i ? $c_r^i(s^r) = d_i \bar{y}_r(s^r)$

Agent i budget constraint: $\sum_t \sum_{s^r} q_r^0(s^r) \underbrace{c_r^i(s^r)}_{d_i \bar{y}_r(s^r)} = \sum_t \sum_{s^r} q_r^0(s^r) y_r^i(s^r)$

$$\Rightarrow d_i = \frac{\sum_t \sum_{s^r} q_r^0(s^r) y_r^i(s^r)}{\sum_t \sum_{s^r} q_r^0(s^r) \bar{y}_r(s^r)}$$